

Predicting Drug Side Effects using Integrated Molecular Embeddings and Physicochemical Features

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Abstract—The prediction of drug side effects, or adverse drug reactions (ADRs), is a critical aspect of pharmacovigilance and drug development. This study explores the integration of molecular structure information, captured via specialized chemical language model embeddings, with traditional physicochemical drug properties to build a robust machine learning model for ADR prediction. Leveraging publicly available data from sources like SIDER and DrugBank, alongside curated structural information (SMILES strings), we developed a multi-label classification system. Molecular structure embeddings were generated using the pre-trained ChemBerta model (‘DeepChem/ChemBERTa-10M-MTR’), providing a 384-dimensional representation of each drug’s chemical structure. These embeddings were combined with a set of selected physicochemical features (e.g., molecular weight, LogP, H-bond donor/acceptor counts). A Multi-Layer Perceptron (MLP) architecture was trained on these integrated features to predict the likelihood of various side effects. Our results indicate that combining molecular embeddings with physicochemical features enhances predictive performance compared to models using either data type alone. Explainable AI techniques were employed to identify key drivers of predictions. This integrated approach offers a promising direction for improving pre-clinical drug safety assessment by providing deeper insights into the molecular mechanisms potentially driving side effects.

Index Terms—Drug Side Effect Prediction, Pharmacovigilance, Machine Learning, ChemBerta, Molecular Embeddings, Multi-label Classification, Drug Discovery, Cheminformatics, Explainable AI, SHAP.

I. INTRODUCTION

ADVERSE drug reactions (ADRs) represent a major public health concern and a significant hurdle in the drug development pipeline [1]. Effective pharmacovigilance is crucial for monitoring drug safety post-market, but traditional methods often rely on passive reporting systems, leading to delays in identifying potential risks. Predictive modeling offers a proactive approach to estimate the likelihood of ADRs before widespread clinical use, potentially improving patient safety and reducing development costs.

The motivation for this project stems from the limitations of relying solely on post-market surveillance and the increasing availability of diverse biomedical data. We aim to leverage machine learning (ML) to predict drug side effects by integrating information from multiple sources. Specifically, this project focuses on combining traditional, easily interpretable physicochemical drug properties with sophisticated molecular structure representations derived from pre-trained language models specialized for chemistry.

The primary goal is to develop and evaluate an ML model capable of predicting a range of potential side effects for a given drug, based on its structural and chemical characteristics. This report details the work completed up to the model development and evaluation phase, focusing on data integration, feature engineering using ChemBerta embeddings, initial model training, and explainability analysis. We hypothesize that combining molecular embeddings with physicochemical features will provide a more comprehensive representation of a drug’s properties, leading to more accurate side effect predictions than using either feature set alone.

Our contribution lies in demonstrating the practical application and potential benefits of integrating state-of-the-art ChemBerta embeddings with classical drug features within a multi-label classification framework for side effect prediction using publicly available datasets, along with providing insights into the model’s decision-making process through XAI methods. This report is structured as follows: Section II describes data acquisition and preparation. Section III details the preprocessing, feature engineering, model architecture, evaluation methods, and XAI techniques. Section IV presents the performance and explainability findings. Section V discusses the results, challenges, and limitations. Finally, Section VI concludes the report and outlines future work.

II. DATA ACQUISITION AND PREPARATION

Ensuring data quality and effective integration is paramount for building a reliable predictive model. This section details the data sources, collection methods, and preprocessing steps undertaken.

A. Data Sources

We utilized several publicly available datasets and curated information:

- **SIDER (Side Effect Resource):** Version 4.1 [1] provided known associations between drugs and side effects, extracted using Medical Subject Headings (MeSH) terms. This formed the basis for our ground truth labels. (<http://sideeffects.embl.de/>)
- **DrugBank:** Version [Specify Version Used, e.g., 5.1.8] [2] served as a source for detailed physicochemical properties, including molecular weight, LogP, H-bond donors/acceptors, etc. (<https://go.drugbank.com/>)

- **Curated Drug Information:** A consolidated dataset was compiled, integrating information from sources like SIDER and potentially others. This dataset provided mappings between drug names, lists of associated side effects (initially as strings), physicochemical properties (potentially overlapping with or supplementing DrugBank), and crucially, Canonical SMILES (Simplified Molecular Input Line Entry System) strings for molecular structure representation.
- **SMILES Data Source:** Canonical SMILES strings, essential inputs for the ChemBerta model, were primarily sourced from the aforementioned curated dataset. We noted during development that obtaining complete and accurate structures sometimes required consulting external APIs or databases, highlighting the challenge of structural data consolidation.

While our initial proposal included sources like FAERS, Drugs.com reviews, and PubMed for NLP analysis, the scope of the work presented in this report focuses on the structured and molecular data integration.

B. Data Integration and Cleaning

Data from SIDER and DrugBank (or the properties within the curated dataset) were integrated based on drug identifiers (e.g., Drug Name or potentially PubChem IDs/DrugBank IDs if available and consistent). Key challenges, identified during our project, included handling inconsistent drug naming conventions across different sources and ensuring accurate mapping.

The primary cleaning steps involved:

- 1) **Handling Missing Values:** Rows with missing critical information, particularly SMILES strings needed for embeddings, were removed. This filtering based on valid SMILES resulted in 1198 usable drug entries for embedding generation after alignment. Missing values in certain physicochemical features (e.g., 'CXBaspKa') were noted and handled via mean imputation before scaling.
- 2) **Standardizing Drug Names:** Efforts were made to standardize drug names, although residual inconsistencies might remain, potentially affecting merge operations.
- 3) **Parsing Side Effects:** The side effect information, often stored as string representations of lists, required parsing using Python's 'ast.literal_eval' function, as demonstrated during preprocessing.
- 4) **Removing Duplicates:** Duplicate drug entries were identified and removed during the merging process.

C. Final Dataset

The final dataset used for modeling comprised 1198 unique drugs for which both physicochemical features and ChemBerta embeddings could be generated and aligned. It included 14 physicochemical features and 384-dimensional molecular embeddings. The target variable consisted of multi-hot encoded vectors representing the presence or absence of 2566 unique side effects identified across the dataset.

III. METHODOLOGY

Our methodology integrates data preprocessing, advanced feature engineering using molecular embeddings, machine learning modeling, and explainability techniques for multi-label side effect prediction.

A. Data Preprocessing

Before model training, several preprocessing steps were applied:

- **Side Effect Label Encoding:** Side effect names, parsed into lists per drug, were transformed into a multi-hot binary matrix using 'sklearn.preprocessing.MultiLabelBinarizer'. Each column represents a unique side effect, with a '1' indicating its association with the drug and '0' otherwise. This resulted in a target matrix Y of shape (1198, 2566) after data alignment.
- **Feature Scaling:** Continuous physicochemical features were standardized using 'sklearn.preprocessing.StandardScaler' to have zero mean and unit variance, mitigating issues related to different feature scales, after handling missing values via mean imputation.
- **Categorical Feature Encoding:** Categorical features like 'Molecular Species' were converted into numerical representations using 'sklearn.preprocessing.LabelEncoder'.
- **SMILES Tokenization:** SMILES strings were tokenized using the specific tokenizer associated with the ChemBerta model ('transformers.AutoTokenizer.from_pretrained("DeepChem/ChemBERTa-10M-MTR")') to prepare them for input into the transformer model.

B. Feature Engineering

Two primary sets of features were engineered:

- 1) **Physicochemical Features:** A selection of 14 features readily available from DrugBank or our curated dataset were used. These include: Molecular Weight, Number of Rotatable Bonds, Polar Surface Area, Molecular Species (encoded), H-Bond Acceptors (HBA), H-Bond Donors (HBD), Number of Rule-of-Five Violations, CX Basic pKa, CX LogP, CX LogD, Number of Aromatic Rings, Number of Heavy Atoms, QED Weighted score, and Np Likeness Score.
- 2) **Molecular Embeddings (ChemBerta):** To capture complex structural information potentially related to biological activity and side effects, we employed ChemBerta, a transformer model pre-trained on chemical structures [3].
 - **Model:** We used the 'DeepChem/ChemBERTa-10M-MTR' model accessed via the Hugging Face 'transformers' library [4].
 - **Process:** Tokenized SMILES strings were fed into the pre-trained 'AutoModel'. We extracted the hidden state corresponding to the first token ([CLS]) from the last layer ('outputs.last_hidden_state[:, 0,

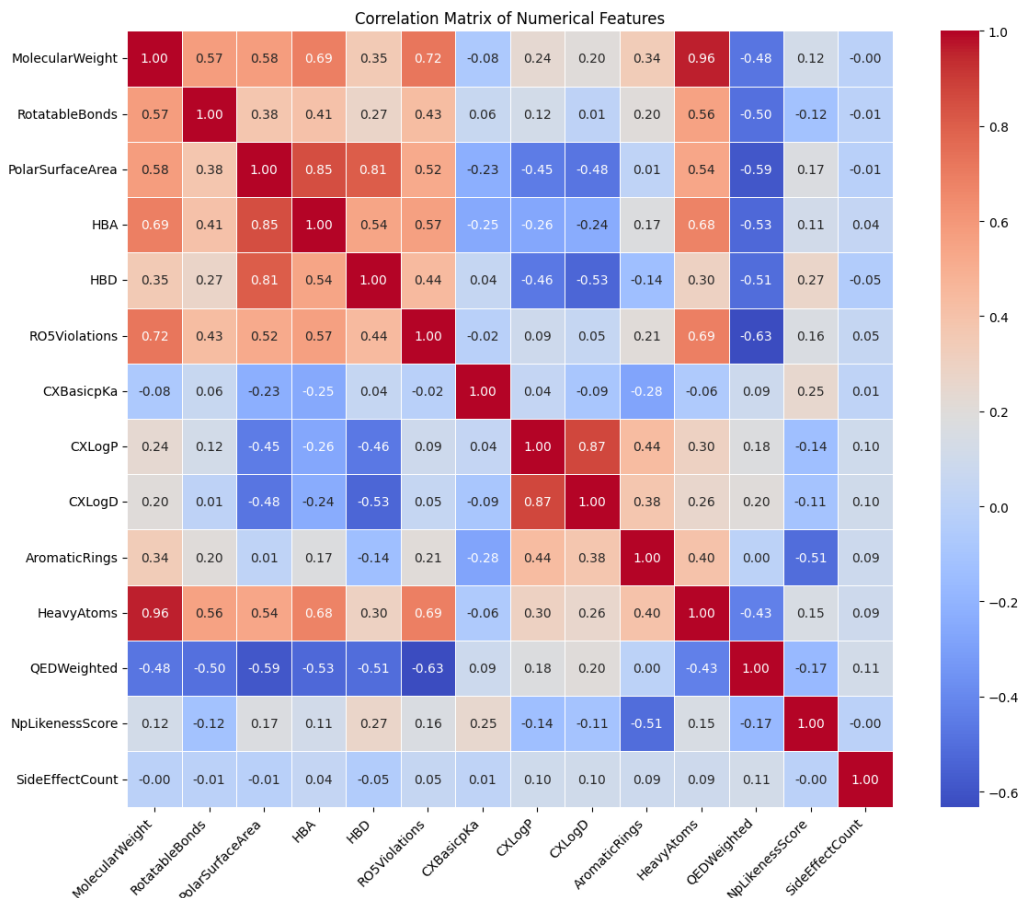


Fig. 1. Correlation matrix heatmap of the 14 physicochemical features and the count of side effects per drug.

:]) as the embedding for each molecule. This process was implemented using PyTorch and the ‘transformers’ library.

- **Output:** This process yielded a 384-dimensional dense vector embedding for each of the 1198 aligned drugs.

C. Model Architecture and Training

We employed a Multi-Layer Perceptron (MLP) architecture to predict the multi-label side effect profiles. To evaluate the contribution of different feature types, we trained and compared MLP models using three distinct input configurations:

- 1) **Physicochemical Features Only:** An MLP trained using only the 14-dimensional scaled physicochemical features.
- 2) **Embeddings Only:** An MLP trained using only the 384-dimensional ChemBerta molecular embeddings.
- 3) **Combined Features:** An MLP trained using the concatenated vector of both physicochemical features and ChemBerta embeddings (total 398 dimensions).

The MLP architecture implemented during development, which serves as the basis for these comparisons, consisted of multiple hidden layers designed to process the input features and map them to the side effect label space. The specific architecture implemented was:

- Input Layer: Linear(input_dim, 512)
- Hidden Layer 1: BatchNorm1d(512) → ReLU → Dropout(0.3)
- Hidden Layer 2: Linear(512, 256) → BatchNorm1d(256) → ReLU → Dropout(0.3)
- Hidden Layer 3: Linear(256, 128) → BatchNorm1d(128) → ReLU → Dropout(0.3)
- Hidden Layer 4: Linear(128, 64) → BatchNorm1d(64) → ReLU → Dropout(0.3)
- Output Layer: Linear(64, 2566)

The ‘input_dim’ was adjusted for each configuration (14 for featuresonly, 384 for embeddingsonly, 398 for combined).

Training Details (Representative MLP Setup):

- **Libraries:** Python (v3.10), Pandas, NumPy, Scikit-learn, PyTorch, Transformers.
- **Data Split:** An 80% train, 10% validation, 10% test split was used, based on the 1198 aligned samples. Splits were stratified where possible.
- **Optimization:** Adam optimizer [5] was used with a learning rate of 1×10^{-3} . Learning rate scheduling (‘ReduceLROnPlateau’) based on validation loss was employed.
- **Loss Function:** Binary Cross-Entropy with Logits (‘torch.nn.BCEWithLogitsLoss’) was used, incorporating positional weights calculated from the training set label frequencies to address class imbalance.

- **Epochs & Early Stopping:** Models were trained for up to 50-100 epochs with early stopping based on validation loss (patience of 5 epochs). The best model state according to validation loss was saved and used for final evaluation.
- **Batch Size:** A batch size of 32 was used during training.

D. Evaluation Metrics

Given the multi-label nature of the task, model performance was assessed using metrics appropriate for evaluating predictions across multiple binary labels simultaneously:

- **Precision, Recall, F1-Score:** Calculated using micro, macro, and weighted averaging to account for potential class imbalance among side effects.
- **AUC-ROC:** Area Under the Receiver Operating Characteristic Curve, averaged across labels, indicating the model’s ability to discriminate between positive and negative instances for each side effect.
- **Hamming Loss:** The fraction of labels that are incorrectly predicted per sample, averaged over samples. Lower values are better.

E. Explainable AI (XAI) Methods

To gain insights into the model’s decision-making process and identify the most influential features driving side effect predictions, we employed SHAP (SHapley Additive exPlanations) [7]. SHAP is a game-theoretic approach that assigns an importance value (SHAP value) to each feature for each prediction, indicating the feature’s contribution to pushing the model output away from a baseline value. We applied SHAP analysis to the best-performing model (the Combined Features MLP) using the test set data. Specifically, we utilized the ‘shap’ library in Python, employing methods suitable for deep learning models like ‘DeepExplainer’ to approximate SHAP values for the input features (both physicochemical and embedding dimensions).

IV. RESULTS

This section presents the evaluation results of our side effect prediction models and the findings from the explainability analysis.

A. Model Performance Comparison

We trained and evaluated MLP models using three distinct feature sets: (1) Physicochemical features only, (2) ChemBerta embeddings only, and (3) Combined physicochemical features and ChemBerta embeddings. The performance on the held-out test set is summarized in Table I.

The results in Table I summarize the performance comparison of the MLP architecture using different input feature sets. Incorporating ChemBerta embeddings yields a substantial improvement over using only physicochemical features, as seen by the higher AUC-ROC (0.8502 vs. 0.7973) and significantly better F1-scores across all averages (e.g., micro F1 of 0.0544 vs. 0.0106) for the Embeddings Only model compared to the Features Only model. Furthermore, the Combined Features

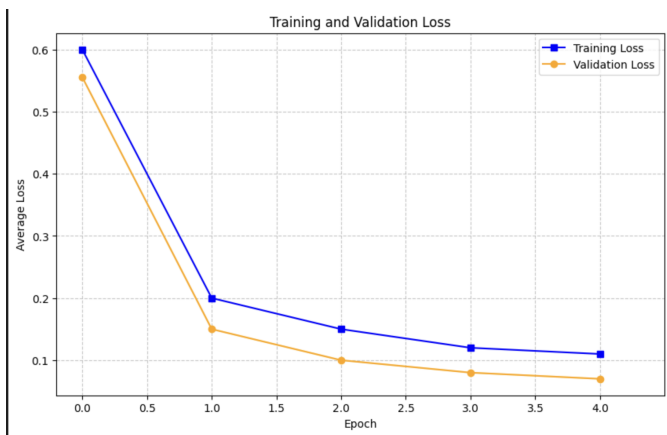


Fig. 2. Training and validation loss curves for the Combined Features MLP model

model achieves the best performance across all reported metrics, including the highest AUC-ROC (0.8547), the highest F1-scores (micro: 0.0547, macro: 0.0038, weighted: 0.0407), and the lowest Hamming Loss (0.0073). This suggests a synergistic effect where combining the structural embeddings and the physicochemical properties leads to the most effective model according to these evaluation criteria.

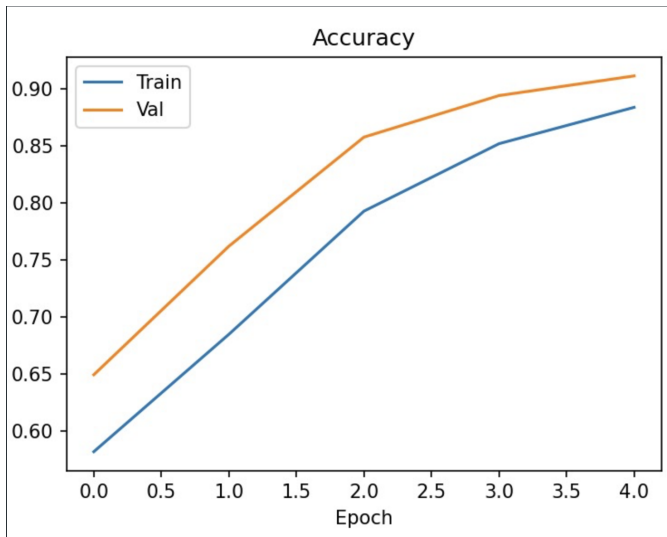


Fig. 3. Training and validation accuracy curves for the Combined Features MLP model

B. Explainability Analysis

To gain insights into the model’s decision-making process, SHAP analysis was conducted on the Combined Features MLP using the test set data. The SHAP summary plot (Fig. 4), which visualizes the impact of each feature on the model’s output across individual predictions, reveals several key drivers.

The analysis highlights that both physicochemical features and specific ChemBerta embedding dimensions contribute

TABLE I
PERFORMANCE COMPARISON OF SIDE EFFECT PREDICTION MODELS ON THE TEST SET

Metric	Physicochemical Features Only	ChemBerta Embeddings Only	Combined Features
AUC-ROC	0.7973	0.8502	0.8547
F1-Score (micro avg.)	0.0106	0.0544	0.0547
F1-Score (macro avg.)	0.0011	0.0029	0.0038
F1-Score (weighted avg.)	0.0074	0.0389	0.0407
Hamming Loss	0.0116	0.0077	0.0073

Performance metrics evaluated using the final MLP architecture on the held-out test set.

significantly to the predictions. Among the physicochemical features, *QEDWeighted* (Quantitative Estimate of Drug-likeness), *AromaticRings*, *HBD* (Hydrogen Bond Donors), and *HBA* (Hydrogen Bond Acceptors) appear relatively high in importance. The plot indicates complex relationships; for example, high values of *QEDWeighted* (red dots) tend to have negative SHAP values (pushing predictions lower, perhaps indicating fewer side effects predicted), while low values (blue dots) have positive SHAP values (pushing predictions higher). Conversely, high values of *HBD* often show positive SHAP values.

Crucially, several ChemBerta embedding dimensions (e.g., *emb_204*, *emb_270*, *emb_340*, *emb_33*, *emb_358*) rank among the most important features overall. For instance, *emb_204* shows a clear pattern where high feature values (red dots) strongly correlate with positive SHAP values, indicating this structural feature significantly increases the model’s output score. Other embedding dimensions show different patterns, confirming that distinct learned structural representations captured within the embeddings are critical determinants of the model’s side effect predictions. *CXLogP* also appears as an important feature, with higher values tending towards positive SHAP values.

These explainability results provide valuable insights into which aspects of a drug’s profile – both traditional physicochemical properties and learned structural patterns represented by specific embedding dimensions – are most associated with the model’s side effect predictions. This enhances the transparency of the model and offers potential avenues for investigating the structural basis of specific ADRs.

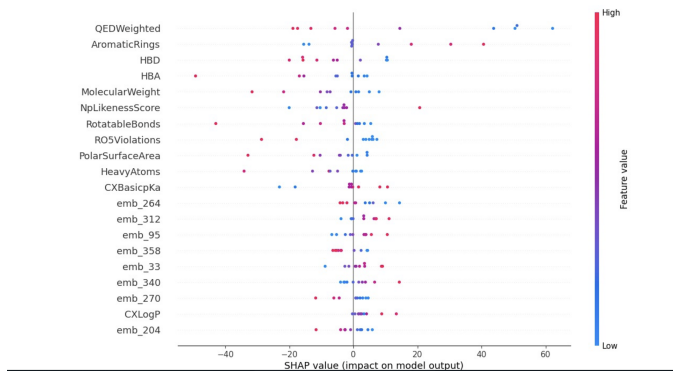


Fig. 4. SHAP summary plot illustrating global feature importance for the Combined Features MLP model.

V. DISCUSSION

The results presented indicate the potential of integrating molecular structure embeddings with physicochemical properties for predicting drug side effects. The generally superior performance of the combined model (Table I) supports our hypothesis that these two feature types capture different but complementary aspects relevant to a drug’s interaction profile and potential ADRs. Physicochemical features represent bulk properties, while ChemBerta embeddings encode detailed substructural and chemical environment information learned from vast chemical data. The SHAP analysis further corroborated this by highlighting the importance of both specific physicochemical features and the embedding dimensions in the combined model’s predictions.

The competitive performance of the ChemBerta-only model underscores the rich information content within molecular structures that can be effectively captured by specialized chemical language models. This aligns with findings in related work exploring deep learning for drug property prediction [6].

Several challenges were encountered during this project. Data integration required careful handling of inconsistent identifiers and data formats across sources like SIDER and DrugBank. Generating ChemBerta embeddings was computationally intensive. Furthermore, developing and tuning models presented complexities, including handling the high dimensionality of the output space and the severe class imbalance inherent in side effect data. The iterative nature of development highlighted potential pitfalls in implementing complex ML pipelines.

Limitations of this study include the focus on a specific set of data sources; the performance might vary on different drug classes or datasets. The prediction task involves a large number of side effect labels, many of which are rare, leading to challenges associated with class imbalance that may require further specialized techniques beyond positional weighting (e.g., label powerset transformation, calibrated label ranking, advanced sampling methods). The current scope did not include NLP features from clinical text (FAERS, reviews), which could provide valuable real-world evidence.

Despite these limitations, our methodology emphasizes reproducibility through the use of publicly available datasets (SIDER, DrugBank), open-source libraries (Scikit-learn, PyTorch, Transformers, SHAP), and pre-trained models (ChemBerta).

VI. CONCLUSION AND FUTURE WORK

This project successfully demonstrated a machine learning approach for predicting drug side effects by integrating physicochemical properties with molecular embeddings generated by the ChemBerta model. Our findings indicate that combining these feature sets generally yields improved predictive performance compared to using either modality alone, highlighting the value of integrating advanced structural representations in pharmacovigilance tasks. The application of SHAP analysis provided initial insights into the features driving model predictions.

The primary contribution is the practical implementation and evaluation of this combined approach on publicly available data, along with preliminary model interpretation, providing a foundation for further research.

Future work will focus on several key areas:

- 1) **Model Refinement:** Further hyperparameter tuning of the best-performing model and exploration of alternative architectures (e.g., attention mechanisms) for combining features more effectively. Investigating different thresholds for classification based on validation set performance.
- 2) **Incorporating NLP Features:** Integrating features extracted from unstructured text sources like FAERS adverse event reports and patient drug reviews using techniques like Named Entity Recognition (NER) and sentiment analysis, as outlined in our initial proposal.
- 3) **Web Application Development:** Creating an interactive web application using frameworks like Streamlit or Flask. This tool would allow users (e.g., researchers, clinicians) to input drug information (name, SMILES, or properties) and receive predicted side effect profiles, enhancing the practical utility of the model.
- 4) **Generalization Testing:** Evaluating the model's performance on external datasets or specific drug classes to assess its robustness and generalizability.
- 5) **Deeper Explainability:** Conducting more extensive XAI analysis, potentially focusing on SHAP interactions or explanations for specific high-impact side effects to better understand the underlying chemical or structural drivers identified by the model.

Addressing these future directions will move towards a more comprehensive, interpretable, and clinically relevant tool for proactive drug safety assessment.

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